WORKED SOLUTION

STEP III 1990, Question 3

Pure • Group Axioms and Conjugation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q3

The key idea

All three parts come from the group axioms alone, and in each one the useful move is to rewrite one element in terms of another. An element of order 2 is its own inverse, which turns (i) into a one-line calculation. In (ii) and (iii) the tool is *conjugation*: $x \mapsto c^{-1}xc$ respects products, so it carries powers to powers.

Throughout we write xy for $x * y$ and e for the identity.

Solution

Part (i)

An element x has order 2 exactly when $x \neq e$ and $x^2 = e$; the second condition says $x^{-1} = x$. So

$$a^{-1} = a, \qquad b^{-1} = b, \qquad (ab)^{-1} = ab.$$

For any group elements, $(ab)^{-1} = b^{-1}a^{-1}$, since $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = e$. Hence

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba,$$

so a and b commute.

Part (ii)

First note that dc is a conjugate of cd:

$$c^{-1}(cd)\,c = (c^{-1}c)\,dc = dc.$$

? **Ask yourself** Why does writing $dc = c^{-1}(cd)c$ help with orders?

Because conjugation passes through powers. For any x and any positive integer n,

$$(c^{-1}xc)^n = (c^{-1}xc)(c^{-1}xc)\cdots(c^{-1}xc) = c^{-1}x^nc,$$

since each adjacent pair $c\,c^{-1}$ in the middle cancels. With $x = cd$ this gives $(dc)^n = c^{-1}(cd)^n c$. Now

$$(dc)^n = e \iff c^{-1}(cd)^n c = e \iff (cd)^n = cec^{-1} = e,$$

where the second step multiplies on the left by c and on the right by c^{-1} (and the converse multiplies by c^{-1} and c). So the positive integers n with $(cd)^n = e$ are exactly those with $(dc)^n = e$. The order is the least such n, so cd and dc have the same order. If there is no such n, both have infinite order, so the statement holds in that case too.

Part (iii)

We are given $c^{-1}bc = b^r$. For a positive integer s, the telescoping argument from (ii) gives

$$c^{-1}b^s c = (c^{-1}bc)^s = (b^r)^s = b^{sr}.$$

The same holds for $s = 0$ (both sides are e). For negative s, write $s = -t$ with $t > 0$; then $c^{-1}b^s c = (c^{-1}b^t c)^{-1}$, because $(c^{-1}b^t c)(c^{-1}b^{-t}c) = e$. So

$$c^{-1}b^s c = (b^{tr})^{-1} = b^{-tr} = b^{sr}.$$

Thus $c^{-1}b^s c = b^{sr}$ for every integer s.

For the second result we use induction on $n \geq 1$, for every integer s at once. The case $n = 1$ is what we have just proved. Suppose $c^{-n}b^s c^n = b^{sr^n}$ for all integers s. Then

$$\begin{aligned} c^{-(n+1)}\, b^s\, c^{n+1} &= c^{-1}(c^{-n}b^s c^n)c \\ &= c^{-1}\, b^{sr^n}\, c \\ &= b^{(sr^n)r} = b^{sr^{n+1}}, \end{aligned}$$

where the last line uses the first result with the integer sr^n in place of s. This completes the induction, so $c^{-n}b^s c^n = b^{sr^n}$ for all positive integers n.

Ask yourself Why did we bother proving the first result for negative s?

If r is negative (for example $c^{-1}bc = b^{-1}$ in a dihedral group), then sr^n can be negative even when s is positive, and the inductive step applies the first result with sr^n in place of s. Proving it for all integers s makes the induction work whatever the sign of r.

Checks

Test on small groups. In the symmetries of a square, the reflections in the two diagonals and their product (a half-turn) all have order 2, and the two reflections do commute, as (i) predicts. In S_3, take $c = (1\,2)$ and $d = (1\,3)$: then cd and dc are the two 3-cycles, both of order 3. For (iii), in the dihedral group a reflection c and rotation b satisfy $c^{-1}bc = b^{-1}$, so $r = -1$ and the formula predicts $c^{-n}b^s c^n = b^{(-1)^n s}$, which is right: conjugating by a reflection twice does nothing. We also checked all three statements by computer, exhaustively on all suitable elements of S_4 (and of S_5 for (i)), and in the group of maps $x \mapsto ux + v$ modulo 13.

Common slips

- Writing $(ab)^{-1} = a^{-1}b^{-1}$. The order reverses; that reversal is exactly what makes (i) work.
- In (ii), proving only that $(cd)^n = e \Rightarrow (dc)^n = e$. You need both directions (or symmetry: swap the roles of c and d), and the infinite-order case should be mentioned.
- In (iii), giving a "proof" by writing three dots for the second result instead of an induction.

Beyond STEP

The map $x \mapsto c^{-1}xc$ is an *automorphism* of the group: a bijection from G to itself that respects the operation, so it preserves orders and sends powers to powers. Part (iii) is the key relation in *semidirect products*: a group generated by b and c with $c^{-1}bc = b^r$ includes the dihedral groups ($r = -1$) and the groups of maps $x \mapsto ux + v$ modulo a prime. Allenby's *Rings, Fields and Groups* develops conjugation and these examples carefully.

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